Innovative Image Dehazing Techniques using Machine Learning Models

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***Abstract*— Image dehazing is a significant task in computer vision, which aims to restore the visibility of images affected by atmospheric particles like fog, haze, and smoke. Traditional model-based dehazing approaches usually depend on prior assumptions for atmospheric light and transmission maps resulting in unsatisfactory estimates for complex scenes. The deep learning techniques particularly the Convolution Neural Networks (CNNs) led to the development of robust and adaptive solutions for image dehazing. This paper provides a survey of CNN based dehazing methods, such as residual networks, a ranking CNN, and multi-scale network. It is based on deep feature extraction and learning-based transmission estimation that can eliminate haze and enhance images. Comparative analyses show that CNN-based methods outperform traditional dehazing approaches in terms of PSNR, SSIM, and visual quality. Experimental results show that deep learning-based dehazing methods are able to effectively dehaze images over various datasets, and such abilities are of great interest for applications in surveillance, autonomous driving, and remote sensing.**

Keywords*— *Image Dehazing, Convolutional Neural Networks (CNN), Deep Learning, Transmission Map Estimation, Residual Networks, Atmospheric Light Estimation, Multi-Scale CNN, Haze Removal

I. Introduction

Image quality diminishes when haze scatters light between objects through atmospheric particles that include dust smoke as well as water droplets thus reducing visibility and degrading contrast and colour fidelity. Different computer vision applications encounter difficulties because of this degradation in their activities which include surveillance, remote sensing, and autonomous navigation. The mainstream methods for dehazing apply enhancements to images while using prior-based physical models to determine light and transmission conditions. The methods struggle to function effectively when prior assumptions stop being valid in sophisticated scenes. Appearance quality degrades because atmospheric dust together with smoke and water droplets scatter light between objects which reduces their visibility on top of corrupting colour accuracy. Computer vision applications face major challenges because of degradations caused by atmospheric elements that reduce visibility during operations including surveillance and remote sensing and autonomous navigation. Standard dehazing technology implements physical prediction systems that measure light and haze transmission although they fail when processing intricate real-world images with nonconforming conditions.

- Importance of Image Dehazing: Image dehazing technology provides essential abilities for computer vision applications that include autonomous vehicles and security systems as well as remote sensing and underwater and photographic imaging. The hazy natural occurrence reduces the accuracy of object detection in self-driving vehicles as well as in security monitoring systems. Haze removal enhances medical analysis particularly in X-ray and microscopic examinations.
- Physics of Haze Formation: The atmosphere creates haze by scattering light rays and absorbing it through small particles in suspension. The scattering process of light through haze depends on Mie Scattering and Rayleigh Scattering processes. Haze gets its definition in ASM as the straight-on illumination from objects with aerial light elements included according to the model.

The essential preprocessing operation known as haze removal helps various computer vision systems to perform object detection tracking and segmentation tasks. Various real-world applications including aerial imagery analysis and medical imaging together with self-driving cars require top-quality image restoration for proper functionality. System performance under adverse environmental conditions becomes more effective when researchers focus on developing solutions for image dehazing which stands as a fundamental research need. The use of deep learning with CNNs has proven to be an effective method for eliminating haze from a solitary image. CNN-based haze removal models excel in dehazing tasks because they extract crucial image features directly from the training data, negating the requirement for transmission maps. The advancement of residual networks, multi-scale CNNs, and ranking-based CNNs has enhanced deep feature extraction in dehazing. These architectural innovations offer improved performance over traditional methods by enabling end-to-end learning and transmission map estimation while reconstructing clear images.

This study explores various CNN-based dehazing techniques and compares them to classical methods, providing empirical evidence regarding their performance capabilities. In the realm of single-image dehazing, deep learning techniques, particularly convolutional neural networks (CNNs), have emerged as the most efficient solution. Data-driven haze removal systems utilizing CNNs can derive essential features from their training datasets without requiring explicit transmission data. The enhancement of dehazing outcomes is attributed to deep feature extraction made possible by residual networks, multi-scale networks, and ranking-based networks. The latest architectural models deliver better outcomes in terms of end-to-end learning, transmission map estimation, and clear image reconstruction when contrasted with conventional techniques. The research paper assesses various CNN methodologies for haze removal and evaluates their performance against traditional techniques through empirical research. Convolutional Neural Networks (CNNs), fueled by advancements in deep learning, have become a powerful tool for addressing single-image dehazing challenges. Data-driven CNN-based models excel at haze removal because they directly acquire features that matter for dehazing during the learning process. The dehazing performance has been improved through different CNN architectures such as residual networks and multi-scale CNNs and ranking-based CNNs. These models achieve end-to-end learning by applying deep feature extraction for estimating transmission maps and reconstructing clear images. Research examines techniques based on CNNs for dehazing alongside conventional methods and presents an evaluation of their performance comparisons. Deep learning and haze density estimation work together to improve visibility together with detailed image preservation in the proposed approach. The findings from experimental tests show that CNN-based dehazing achieves better results than traditional methods regarding perceptual quality and PSNR and SSIM measurements.

II. Literature Review

Computational vision specialists have been trying to address image dehazing problems since early times through physical model and human-designed prior approaches for transmission map and atmospheric light estimation. The traditional methods fail to deliver satisfactory outcomes because real-world conditions break the assumptions about atmospheric light uniformity. The development of deep learning allowed Convolutional Neural Networks (CNNs) to take advantage of data-driven learning algorithms that produce advanced automated and robust dehazing solutions.

A. Traditional Approaches to Image Dehazing

The conventional dehazing methodology depends on the atmospheric scattering model to analyse hazy images as they contain both the original scene radiance and illumination from atmospheric particles. Dark Channel Prior (DCP)[1] and Colour Attenuation Prior and Guided Filtering together with other recent methods represent popular methods for estimating transmission maps as in Fig. 1. The dehazing techniques normally work successfully but their effectiveness diminishes significantly when hazy scenes demonstrate varied depths or bright areas in the sky. Traditional image dehazing procedures split hazy images into two parts through the atmospheric scattering model to obtain scene radiance and atmospheric light. As per the Dark Channel Prior (DCP) approach transmission maps are estimated through the assumption that haze-free images contain one pixel with minimum colour intensity. The performance of DCP deteriorates in both heavily hazy circumstances and when hazy conditions include bright blue skies.

The supervised learning method CAP applies depth modelling using pixel intensity relationships in hazy images. The improvements made by CAP over DCP remain limited because it demonstrates inaccurate depth results in demanding lighting situations. Edge preservation in transmission maps becomes improved through systematic filtering techniques and heuristic-based methods. Several hazy image restoration methods show moderate results but they cannot adequately address actual pictures which display various depth levels together with multiple lighting situations and irregular haze patterns. These methods succeed in certain dehazing applications but fail to deliver satisfactory results on real-world images affected by diverse conditions including depth variations and uneven dispersion of haze across the scene.

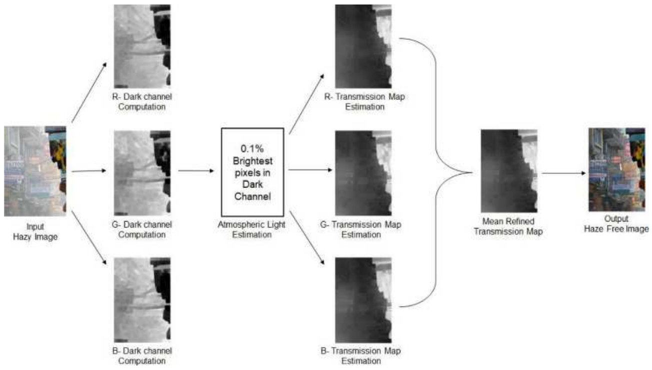

Fig. 1. Dark Channel Prior (DCP) Model

B. Hybrid and Specialized CNN Models

Scientific research has produced hybrid deep learning models which unite deep learning methods with physical knowledge of the subject. The Parallax Attention Mechanisms (PAM) along with Graph-Based CNNs achieved better transmission estimation performance through their capacity to handle geometric and pixel-based information simultaneously. The utilization of self-guided progressive feature fusion allows reference images to boost the dehazing quality by integrating into the CNN chain [2-4]. Recent investigations have established deep learning methods that integrate physical framework with deep learning technology including: PAM improves transmission estimation through an integrated system combining both geometric information and pixel-based information. The technique uses reference images to increase dehazing quality through self-guided progressive feature fusion. The method uses graph networks to enhance haze estimation and improve edge detection as part of its framework.

C. CNN-Based Approaches for Image Dehazing

Modern advancements in deep learning technology have produced CNN-based dehazing models which automatically learn transmission information and dehazed output from image data. Some key architectures include: The improved version of Cycle GAN produces high-quality dehazed images through its ability to perform accurately across multiple datasets without needing matched training data[5]. The dehazing performance of Multi-Scale CNNs and Perceptual Pyramid Networks increases due to their ability to teach features at various scales which preserves edges and reconstructs details[6-10]. The practice of residual learning enables improved transmission estimation and clearer image results by teaching models to recognize hazy and clear pattern differences. End-to-end CNN model-based methods combine encoder-decoder structures for direct conversion of hazy images into haze-free results thus reducing the need for transmission mapping estimations. Image dehazing has experienced a revolutionary change through deep learning which delivers data-intensive solutions. Some notable CNN-based architectures include: Multi-scale CNNs serve different resolution levels in order to capture large-scale and small-scale image features, which both retain details even when removing haze. The networks leverage residual connections to boost their extraction of features and production of haze-free images in frameworks named Residual Learning-Based CNNs. The Cycle GAN-based approaches utilize adversarial training to produce high-quality dehazed images even when operating without paired training samples. End-to-End CNN

Models function through encoder-decoder systems for turning hazy imagery directly into haze-free results without requiring transmission map identification.

While CNN-based dehazing approaches succeed they encounter three main problems including unacceptable color distortions and degraded fine details and expensive computational requirements. Researchers are conducting future work on self-supervised and lightweight systems along with domain adaptation solutions that enhance both performance speed and reliability [3]. The research explores image dehazing methods starting from traditional prior-based approaches and ending with deep learning through CNNs which leads to better performance outcomes. The CNN-based dehazing models experience the following problems even while successfully performing the task: High-definition details disappear along with hue modifications occurring in the image. Real-time application remains challenging because these models require high computational powers. The transfer of knowledge from synthetic datasets to real-world images emerges as a main issue since generalization becomes problematic. The challenges preventing success with CNN-based dehazing models include: Both colour distortions and delicate details become lost from the image. Real-time application of the models becomes restricted by their high computational requirements. The generalization capability of models becomes a problem since they learn from synthetic data but show limitations when processing real images. The field looks into three approaches of self-supervised learning together with lightweight CNN architectures as well as domain adaptation methods to boost efficiency and robustness. Self-supervised learning parallel to lightweight CNN architecture developments and domain adaptation techniques drive current research efforts to enhance robustness and efficiency. Other relevant technologies used are documented in the following research papers [10-24].

III. Research Methodology

The system design for image dehazing using CNNs includes stepwise operations which start with preprocessing then move to network selection followed by training and evaluation.

A. Data Preprocessing

The methodology selects datasets from NTIRE2018-Dehazing together with RESIDE and NYU Depth Dataset for collecting synthetic and real-world images. The technique enhances dataset diversity by applying several data augmentation methods including contrast enhancement along with rotation and flipping and random cropping and image rotation. The normalization process applies to image inputs to enhance the training speed. The training and evaluation of CNN-based dehazing models make use of NTIRE2018-Dehazing, RESIDE, and NYU Depth Dataset. The model's generalization is enhanced through data augmentation methods which include contrast enhancement and rotation together with flipping and random cropping.

B. Network Architecture Solutions

Different CNN architectures are explored:

- Multiple Scale CNNs obtain refined dehazing results by performing feature extraction on various levels.
- Residual-Based CNNs achieves better hazy conditions removal by incorporating residual learning as a core functionality.
- A cascaded network architecture contains succession of subnetworks that conduct progressive improvements.
- Self-Guided CNNs with Progressive Feature Fusion: Integrates reference images for enhanced dehazing.
- Pixel-Guided CNN with PAM: Employs attention mechanisms and graph-cut optimization for precise haze removal.
- Different versions of CNN have been researched for image dehazing work because they present distinct advantages during operation.

C. Training Process

Each dehazing model usually employs one of these loss functions during its operation. The MSE metric produces a pixel-level measurement which evaluates the difference between dehazed results and reference ground truth outputs. Perceptual Loss ensures the dehazed image preserves the same structural pattern as observed in real images [7].

- Loss Functions: MSE, Perceptual Loss, and Adversarial Loss for enhanced realism. Adam optimizer together with batch normalization and dropout and learning rate scheduling represent optimization techniques in this approach.
- Adversarial Loss (for GAN-based models): Encourages realistic haze-free images.
- Optimization Techniques: Training relies on Adam optimizer with its adaptive learning rate mechanism. The model uses batch normalization alongside dropout techniques because these methods help stop overfitting issues from occurring.

D. Model Evaluation

- Quantitative Metrics: Two quantitative indicators PSNR and SSIM assess the image clarity of dehazed images. FSIM calculates perceptual quality through its evaluation process. Without requiring ground truth BRISQUE evaluates image quality as one of the No-Reference Metrics.
- Qualitative Analysis: The testing procedure includes visual assessments with advanced dehazing techniques for establishing actual operational effectiveness.
- PSNR & SSIM: Measures image clarity and structural similarity.
- FSIM & BRISQUE: Evaluates perceptual quality and no-reference image assessment.

E. Advanced Optimization Techniques

Image dehazing performance from CNN models depends on optimized strategies which enhance network performance and optimize accuracy as well as decrease computational requirements. The optimization strategies adopt superior training approaches in combination with parameter fine-tuning and unorthodox learning rules to improve CNN operation. Academic research works on building new approaches which enable CNNs to process dehazing operations at high quality levels with real-time processing requirements.

- Meta-Learning for Improved Adaptability: A primary issue in using CNN for dehazing arises from dataset-specific training because the models show reduced effectiveness on various types of hazes outside their original parameters. The learning paradigm of meta-learning helps dehazing networks to adjust to new haze situation while minimizing the necessity for complete network retraining. The meta-model receives training with numerous hazes to adapt effectively to previously unseen conditions while requiring limited new data. Custom meta-learning techniques for image dehazing have brought benefits in aerial imaging because haze patterns differ widely across town and countryside regions. The utilization of few-shot learning enables CNN models to achieve quick adjustments in different haze conditions automatically with minimal need for large datasets during real-time operations. Such capabilities of meta-learning enable its practical application in satellite imaging along with autonomous driving systems and real-time surveillance tasks.

- Hyperparameter Optimization for CNNs: The execution of CNN-based dehazing models strongly depends on various hyperparameters that include learning rate and batch size alongside dropout rate and convolution kernel size. Achieving optimal setting of these parameters enables networks to converge effectively while avoiding both underfitting and overfitting issues.

Learning Rate Adjustment: Using a high learning rate can destabilize the learning process, while a low learning rate may result in excessively slow convergence. Adjusting the learning rate through Cyclic Learning Rate (CLR) combined with Exponential Decay during the training phase yields the best model results.
Dropout and Regularization Techniques: Training CNNs for dehazing often leads to overfitting, as they can overly memorize the haze patterns found in the training dataset. By employing dropout layers that temporarily deactivate certain neurons, the models can reduce their dependency on specific features, enabling the dehazing algorithms to generalize more effectively.
Neural Architecture Exploration (NAE): Through an automated approach, NAE investigates suitable combinations of CNN architectural components, including the selection of layers, activation functions, and kernel sizes. This method allows researchers to identify efficient CNN configurations for haze removal that operate with reduced computational demands.

- Optimization Techniques for CNN Training: To develop accurate dehazing CNN models with effective optimization methods is vital due to the need to manage computational costs. Common optimization strategies include:

Adam Optimizer: Adam (Adaptive Moment Estimation) is widely used due to its integration of momentum updating with adaptive learning rate strategies. The implementation of CNN models benefits from the Adam optimizer, allowing for rapid convergence while accommodating varying levels of haze complexity.
Stochastic Gradient Descent (SGD) with Momentum: The weight adjustment process with SGD becomes more stable as it incorporates momentum, helping to mitigate fluctuations in training loss. This technique is particularly advantageous for detailed dehazing tasks.
Adaptive Gradient Clipping (AGC): AGC safeguards dense dehazing models from unstable training by regulating potentially harmful weight update sizes.

- Efficient CNN Architectures for Real-Time Dehazing: The execution of real-time dehazing functions on smartphones, drones, and IoT cameras relies on resource-efficient CNN architectures. To facilitate this, researchers employ: Model quantization, which converts floating-point weight representations in CNNs to lower-memory 8-bit formats, also enhances the speed of inference operations. Network pruning reduces unnecessary neurons and filters while lowering computational demands, all while preserving dehazing performance at a comparable level. This results in tiered dehazing effectiveness with minimal energy consumption, making them suitable for embedded AI applications. The MobileNet and EfficientNet variants represent effective CNNs that deliver top-tier dehazing outcomes with minimal energy requirements which makes them appropriate for embedded AI implementations. Researchers make dehazing models faster and more accurate for practical applications by implementing meta-learning together with hyperparameter optimization along with efficient optimization algorithms and lightweight convolutional neural network designs. Quantum computing together with federated learning. getZ will evolve CNN-based dehazing because they help decrease training duration and expand operational capabilities.

IV. Ethical Considerations And Bias In Image Dehazing

CNN-based image dehazing approaches in real-world implementations require immediate solutions for their associated ethical problems and biases. The main ethical concerns about AI-enhanced dehazing systems refer to fairness and transparency along with privacy and possible misuses.

- Fairness and Bias in CNN-Based Dehazing: The main ethical challenge when utilizing AI-based dehazing arises from the presence of bias in training data sets. The majority of dehazing model training utilizes synthetic haze datasets along with images obtained from particular geographical areas that might not align with global atmospheric conditions. The dehazing models developed for urban haze training environments show low generalization ability when used for dehazing rural places or coastal areas because their atmospheric conditions differ substantially. A comparable level of performance is observed under different real-world scenarios, which raises safety concerns when implementing essential applications like autonomous driving and surveillance. Certain CNN-based dehazing techniques can introduce color inaccuracies in particular contexts, leading to misleading images that may confuse users. Using dehazed images improperly in legal or forensic inquiries or intelligence operations could lead to biased and unfair outcomes due to flawed interpretations.

- Privacy Concerns in AI-Enhanced Vision Systems: AI-driven dehazing technology employed in surveillance cameras raises growing privacy issues due to its implementation. The improved visibility provided by dehazing technology generates ethical concerns regarding the storage, processing, and use of AI-enhanced surveillance data. The analysis of facial characteristics and individual tracking activities, alongside surveillance of restricted areas, becomes feasible through dehazing technology utilized by governmental law enforcement agencies. The improper deployment of these systems poses a threat to personal privacy via widespread surveillance techniques. Cascade CNN models can lead to unethical applications when not regulated, as they may eliminate digital censorship from images, enabling unauthorized data access and the development of deepfake technology. CNN-based dehazing models exhibit significant transparency issues, which constitute a major ethical concern. The decision-making processes within deep learning models function as opaque mechanisms that are not easily understood by humans. Identifying errors in autonomous operations or military surveillance systems due to unexplained dehazing challenges can result in serious operational implications. Research teams are concentrating on developing explainable AI (XAI) solutions to enhance CNN-based dehazing techniques, promoting improved transparency, interpretability, and accountability. To address these ethical challenges, researchers and policymakers should collaborate on three key areas. There is a need for new research to create unbiased datasets that encompass diverse environmental conditions for CNN-based dehazing systems, ensuring fair global performance. Organizations must implement frameworks to regulate the ethical conduct of AI vision systems. Researchers ought to employ interpretable CNN architectures to enhance model explainability, thereby decreasing image-processing mistakes caused by unintended errors. The implementation of privacy protection strategies should be integrated into AI-based surveillance systems to stop unauthorized data exploitation. Researchers must focus on ethical elements to develop transparent AI-based dehazing solutions that reduce concerns resulting from technology bias and violations of privacy and misuse of AI technologies as in Table 1.

TABLE I. COMPARISON OF IMAGE DEHAZING METHODS USING CNNS ACROSS COUNTRIES

Method	Country	Key Features	Advantages	Limitations
Pixel-Guided CNN with Parallax Attention Mechanism (PAM) via Graph Cut	Saudi Arabia	Uses pixel variance and parallax attention to refine transmission maps	More detail extraction, better visibility improvement	Over-reliance on real ground truth images
Deep Generative Adversarial Network (GAN) for Image Dehazing	India	Uses perceptual loss functions to enhance high-level feature extraction	Avoids pixel-wise loss, works well on various scenes	Requires large datasets for training
Multi-scale CNN with Holistic Edges	China & USA	Coarse and fine-scale networks refine transmission maps using holistic edges	Faster and more efficient, preserves edge information well	Struggles with high haze density in some images
Residual-Based Deep CNN	China	Uses a two-phase approach: transmission map estimation and residual learning	Avoids explicit estimation of atmospheric light, improves efficiency	Can introduce slight color distortions
Cascaded CNN for Joint Transmission & Atmospheric Light Estimation	China & Australia	Jointly estimates medium transmission and global atmospheric light using cascaded subnetworks	More accurate and robust dehazing	Amplifies noise in certain images
Perceptual Pyramid Deep Network	USA	Multi-scale encoder-decoder structure with dense and residual blocks	Better contextual understanding of haze distribution	High computational cost

V. INDUSTRY USE CASES

Various industries use wide applications of CNN-based dehazing methods to boost visibility while improving accuracy in different domains.

- Autonomous Vehicles & Transportation: Self-driving cars achieve enhanced real-time object detection during autonomous operation due to the elimination of hazy conditions. Used in traffic monitoring systems for better visibility in foggy or rainy conditions. Enhances real-time object detection in foggy conditions.
- Surveillance & Security: Surveillance CCTV cameras benefit from the methodology which enhances image visibility when operating in poor weather to achieve improved object recognition. Used by law enforcement agencies for forensic image enhancement. Improves forensic analysis and object recognition.
- Aerial and Satellite Imaging: The technology helps satellite and UAV images gain better image clarity during remote sensing operations. Enhances visibility for remote sensing applications. Emergency response teams use digital image enhancement to obtain better visibility when responding to natural disaster areas including wildfires, floods, storms.
- Healthcare & Medical Imaging: Application in medical imaging technology and microscopy enhances observation of unclear or foggy medical scans. Removes artifacts in microscopic images.
- Consumer Photography & Entertainment: Integrated into smartphone cameras for automatic haze removal in outdoor photography. Through its deployment the film and gaming sectors achieve better visibility in scenes surrounded by fog. Automates haze removal in smartphone cameras.
- Underwater & Maritime Imaging: Underwater exploration teams and maritime navigators employ the technology to eliminate distorting haze from water particles. Improves underwater visibility for marine exploration.

VI. INDUSTRY CASE STUDIES

Various industries now use CNN-based image dehazing at a high rate to ensure clear and unobstructed images because they are crucial for operational accuracy and user safety. Real-world applications of deep learning-based dehazing demonstrate their success at clearing up images throughout different sectors including automotive automation and space observation systems and video monitoring and medical picture analysis.

- Google's AI for Weather Imaging: The investment of Google in artificial intelligence technology has led to enhanced satellite image clarity which finds its primary application in weather prediction and environmental surveillance tasks. Satellite images become difficult to evaluate properly because of atmospheric haze together with clouds and environmental pollutants blocking the visibility. Google AI systems use CNN-based dehazing approaches to generate clear and high-definition surface images of Earth which removes atmospheric distortions produced by haze elements and airborne particles. This technology finds its main practical use for disaster monitoring purposes. The act of natural disasters including fires and hurricanes and floods together with hazy environmental conditions obstruct satellite imaging observations leading to extended response durations. The combination of multi-scale CNNs alongside GAN-based dehazing models operates in Google's AI systems to improve image clarity thus enabling emergency response teams to obtain precise real-time visual data. The dehazed images function with geospatial AI to measure vegetation changes as well as regional development patterns and environmental climate shifts which makes this technology highly valuable for ecological research.

- Tesla's Autonomous Driving System: Self-driving vehicles rely on computer-operated vision systems to interpret road conditions and detect obstacles to ensure their safety. Due to adverse weather conditions such as fog, haze, and rain, vehicle cameras capture images that lack clarity. Tesla's vision system incorporates CNN-based image dehazing technology as part of its self-driving features to enhance the detection of objects during challenging weather situations. Tesla Autopilot employs deep residual CNNs that improve the quality of low-resolution images captured in foggy or hazy environments. These CNN networks analyze images at different depths to distinguish between actual objects and the effects of haze. Through self-supervised learning, Tesla's AI enhances its dehazing capabilities, which leads to improved detection accuracy over time. Dehazed images are crucial to Tesla's sensor fusion system, as they integrate camera data with inputs from radar and LiDAR to create a more advanced understanding of the scene. Autonomous driving systems effectively detect lanes, pedestrians, and avoid accidents due to precise hardware performance, even in weather conditions that negatively affect visibility. The CNN-based dehazing approach utilized by Tesla contributes to enhanced road safety by providing clear visual information that improves the reliability of autonomous vehicles.

- Military and Security Applications: Military organizations and security agencies utilize CNN-based dehazing techniques since they enhance the quality of surveillance videos in challenging environmental circumstances. Security personnel at borders and law enforcement adopt dehazing methods that enable better object observation through fog, darkness, and smoke. Military drones equipped with thermal and infrared cameras experience haze distortions that hinder their ability to identify potential threats or targets. Better actionable military intelligence is gained through improved surveillance technology, as deep learning methods like GAN-enhanced techniques paired with attention-based CNNs refine aerial surveillance video. Security applications gain significantly from CNN-based dehazing due to its image enhancement capabilities. Multi-task CNN architectures empower investigators to recover crime scene footage, even if recorded under poor lighting conditions with smoke or fog, thereby uncovering critical details such as facial features, vehicle license plates, and suspicious activities.

- Aerial and Satellite Imaging: The distortions caused by haze in satellite images from agencies NASA and ESA obstruct the validity of these images for performing land classification and conducting agriculture monitoring and urban planning. The combination of multi-scale CNNs with dedicated edge-preserving tools allows satellite data enhancement to produce better results for geographical and environmental assessment. The process of clearing haze improves researchers' detection abilities in identifying deforestation patterns as well as measuring glacier melting rates while monitoring land-use changes for climate change research.

- Medical Imaging and Healthcare: Medical professionals have utilized dehazing methods to enhance imaging procedures in microscopy and X-ray diagnostics. The presence of hazy distortions obscure vital information within medical images acquired under low-light or biological fluid transmission. Frames processed by CNN-based dehazing systems become clearer so radiologists together with pathologists can better locate medical abnormalities. The implementation of CNN-based dehazing technology brought revolutionary changes across many sectors because it improves image clarity as well as supports better decision-making while enhancing the accuracy of essential applications.

VII. Shortcomings And Challenges

Multiple hurdles continue to affect the effectiveness of CNN-based methods used for image dehazing although their results have been successful. Numerous CNN models produce erroneous color restorations which results in unnatural hues throughout dehazed outputs. The haze removal process through deep learning approaches includes two major challenges because some models either make images too vibrant or leave haze remaining within complex images that display non-homogeneous fog effects [6]. The high complexity of deep CNN architectures creates obstacles for real-time dehazing operations because edge devices and low-power applications struggle to process such complex systems. Real-world hazy images from various scenes fail to perform well with synthetic dataset-trained models because they contain different haze properties and lighting along with varying depths between objects. The requirement for large datasets during network training becomes challenging because obtaining suitable real-world hazy condition datasets is difficult. Edge Artefacts along with Detail Diminution become issues among some CNN-based approaches because of their processing of high-density haze areas. An array of issues hampers the deployment of CNN-based dehazing techniques after considering their notable strengths. The presence of heavy fogginess in images leads several models to create abnormally coloured results along with visible artifacts. Deep CNN architectures need substantial processing power which limits their ability to do real-time dehazing applications. Dehazing models trained with synthetic haze patterns do not effectively work on actual world images containing diverse levels of density in the hazy areas. Real-world hazy image datasets of good quality and substantial size represent a major barrier to effective training and evaluation of hazy image models. Deep learning-based dehazing requires training datasets and computational resources that prove to be costly for developing new models. The edge detail preservation along with the loss of fine details becomes a challenge for CNN-based methods working in dense haze regions. Researchers should resolve these difficulties through the creation of effective architectures and improvements to dataset quality and increased model universal applicability capabilities.

CONCLUSION

Image dehazing forms an essential component of computer vision which finds practical uses in autonomous driving and remote sensing and surveillance and photography industries. The introduction of CNN-based dehazing techniques in haze removal has outperformed traditional methods because they learn deep hierarchical features. The solution faces three main drawbacks that include color distortions and challenges in computational complexity and generalization difficulties. Further studies need to concentrate on creating hybrid dehazing models along with self-supervised learning systems and real-time procedures to enhance the practicality and speed of dehazing methods based on CNNs. The development of deep learning technologies alongside attention mechanism capabilities and better computational systems will drive image dehazing improvements which reduce the difference between artificial datasets and practical implementations. The performance of CNN-based image dehazing surpasses traditional methods because of its ability to use deep hierarchical features. Future investigations need to develop combination dehazing models and real-time dehazing solutions and improved generalization methods to close the synthetic dataset-application divide.

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